

Generative AI Engineer Roadmap

A Complete Step-by-Step Guide from Beginner to Production-Ready AI Architect

If your goal is to become a **Generative AI Engineer** who can build production-ready AI applications (not just use ChatGPT), here is the complete roadmap from beginner to advanced.

Recommended Timeline

Beginner	Intermediate	Advanced	Production
0–2 Months	2–5 Months	5–8 Months	8–12 Months
Python, Math, ML, Git, SQL	Deep Learning, NLP, Transformers, LLMs, Prompt Eng.	RAG, Vector DBs, AI Agents, MCP, FastAPI, Docker	Fine-Tuning, LLMOps, Security, Multimodal, Architecture

The Core Learning Phases

Phase 1: Programming Fundamentals (2–4 Weeks)

LEARN PYTHON

- Variables, Data types & Loops
- Functions & Object-Oriented Programming (OOP)
- Exception & File handling
- Virtual environments & pip

LIBRARIES & PROJECTS

numpy pandas matplotlib

- **Projects:** CSV Analyzer, Calculator, Student Management System

Phase 2: Mathematics (3–4 Weeks)

CORE MATH AREAS

- **Linear Algebra:** Vectors, Matrices, Dot Product, Matrix Multiplication
- **Probability:** Mean, Variance, Bayes Theorem
- **Calculus:** Derivatives, Gradient Descent

RECOMMENDED RESOURCES

- 3Blue1Brown (Essence of Linear Algebra / Calculus)
- Khan Academy (Statistics and Probability)

Phase 3: Machine Learning (1 Month)

CONCEPTS & ALGORITHMS

- Supervised vs. Unsupervised Learning
- Classification, Regression & Clustering
- Linear & Logistic Regression, Decision Trees
- Random Forest, SVM, KNN

LIBRARY & PROJECTS

scikit-learn

- **Projects:** House Price Prediction, Spam Detection, Customer Churn Analysis

Phase 4: Deep Learning (1 Month)

CORE ARCHITECTURE

- Neural Networks, Activation & Loss Functions
- Optimizers (Adam, SGD)
- CNN (Computer Vision), RNN & LSTM

FRAMEWORK & PROJECTS

PyTorch

- **Projects:** Digit Recognition (MNIST), Image Classification

Phase 5 & 6: NLP Basics & Transformers (4–5 Weeks)

NLP ESSENTIALS

NLTK spaCy

- Tokenization & Word Embeddings
- Word2Vec & TF-IDF

TRANSFORMER ARCHITECTURE

Hugging Face Transformers

- Encoder, Decoder, Self-Attention, Multi-head Attention, Positional Encoding
- **Reading:** "Attention Is All You Need" Paper

Phase 7 & 8: LLMs & Prompt Engineering

LLM FUNDAMENTALS

- GPT, Llama, Mistral, Gemma, Qwen, Claude
- Context Window, Tokens, Temperature, Top-p
- Fine-tuning & Quantization basics

PROMPT ENGINEERING

- Zero-shot, Few-shot, Chain of Thought (CoT), Role prompting, XML / JSON structured outputs
- **Projects:** AI Tutor, AI Resume Reviewer, AI Interviewer

Phase 9, 10 & 11: Embeddings, Vector DBs & RAG

VECTOR MANAGEMENT

- Vector embeddings, Cosine similarity, Similarity search
- **Models:** BGE, E5, OpenAI Embeddings, Sentence Transformers
- **Databases:** FAISS, Chroma, Pinecone, Weaviate, Milvus, Qdrant

RETRIEVAL-AUGMENTED GENERATION

LangChain LlamaIndex

- Document Loading, Chunking, Retrieval, Re-ranking
- **Projects:** Chat with PDFs, Company Knowledge Assistant, Legal Assistant

Phase 12 & 13: AI Agents & Model Context Protocol (MCP)

AUTONOMOUS AGENTS

LangGraph CrewAI AutoGen

- Agent Loop, Planning, Memory, Function / Tool Calling
- **Projects:** Research Agent, Coding Agent, Email Assistant

MCP ECOSYSTEM

- MCP Server & Client architecture, Tool registration
- **Projects:** Database MCP, GitHub MCP, Frappe MCP, Gmail MCP

Phase 14 & 15: Fine-Tuning & AI Evaluation

FINE-TUNING TECHNIQUES

PEFT TRL Unsloth

- LoRA, QLoRA, Parameter-Efficient Fine-Tuning
- SFT (Supervised Fine-Tuning), DPO, RLHF (conceptual)

EVALUATION FRAMEWORKS

Ragas DeepEval LangSmith

- Hallucination, Faithfulness, Groundedness metrics
- Latency & Token Cost tracking

Phase 16, 17 & 18: Backend APIs, Databases & Deployment

API & DATA ARCHITECTURE

- **FastAPI:** REST APIs, Authentication, OAuth, WebSockets
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis (Caching)

HOSTING & DOCKER

- Docker, Kubernetes (basics), Nginx
- Hugging Face Spaces, Render, Railway, AWS, Azure, GCP
- **Projects:** AI API, Chat API

Phase 19, 20 & 21: MLOps/LLMOps, Production Systems & Security

PRODUCTION SYSTEMS

MLflow Weights & Biases DVC

- Streaming, Async programming, Celery, RabbitMQ, Kafka
- Rate limiting, Monitoring, Logging, Tracing

AI SECURITY

- Prompt Injection & Jailbreak mitigation
- Secrets Management, Authentication, Authorization

Phase 22 & 23: Open Source LLMs & Multimodal AI

LOCAL MODELS

Ollama vLLM LM Studio

- Run Llama, Mistral, Gemma, Qwen, Phi locally

MULTIMODAL AI

Whisper YOLO SAM Diffusers

- Vision Models, OCR, Image Gen, STT, TTS
- **Projects:** Image Chatbot, OCR Assistant

Suggested Learning Order Summary

Following this layout ensures your core fundamentals are locked down before addressing complex abstractions:

1. Python Programming

2. Mathematics Foundations

3. Machine Learning Concepts

4. Deep Learning Essentials

5. Natural Language Processing (NLP)

6. Transformer Architectures

7. LLM Fundamentals & Tokenomics

8. Advanced Prompt Engineering

9. Vector Embeddings

10. Vector Databases

11. RAG Architecture

12. AI Agents & Frameworks
1. Model Context Protocol (MCP)

2. Fine-Tuning Methodologies (LoRA/QLoRA)

3. AI System Evaluation & Metrics

4. Backend Application Deployment

5. Production Web APIs

6. Relational & NoSQL Databases

7. MLOps & LLMOps Lifecycles

8. AI Application Security

9. Multimodal AI Models

10. Production Scale & Infrastructure

11. End-to-End Portfolio Development

Phase 24: Elite Portfolio Projects

- ChatGPT Clone

• PDF Chatbot / RAG

• AI Code Reviewer

• AI Resume Analyzer

• AI Interview Coach
- Multi-Agent System

• Voice & Vision Assistants

• RAG over SQL Database

• AI Email Assistant

• AI Customer Support Bot
- AI Document Analyzer

• AI CRM Assistant

• AI ERP Assistant

• AI Finance Assistant

• AI Research Assistant

Phase 25: Essential AI Ecosystem Tools

- Git & GitHub

Linux OS

Docker Containers

VS Code / Jupyter

Postman API

Cursor & Windsurf
- Claude Code

Ollama Environment